

WAGER: Within-cell Antisymmetric Gain Evaluation of Reasoning

Abstract

Vision benchmarks with strong annotation priors can reward a new model whose gain does not come from better instance-specific visual prediction. Existing audits compare a model against a frequency baseline, confounding how well each system fits the prior with whether the *difference* between them uses information beyond it. We introduce WAGER, a pairwise attribution algorithm. For two frozen probabilistic models it forms their proper-score contrast, matches examples sharing a declared prior feature and crosses their labels, giving an exact finite-sample identity: observed gain equals a prior-transported gain plus an antisymmetric instance-alignment gain. The residual is an order-two U-statistic which, under the quadratic score, equals twice the within-cell prediction-label covariance gained by the new model, linking it to the classical resolution term of proper-score decomposition. It costs $O(NK)$ time, needs no fitted prior model, and admits cluster-robust inference. Two further exact results follow: leave-one-out transport removes an attenuation bias of $(n_c - 1)/n_c$ that a naive in-sample plug-in incurs, and coarsening the audit cell shifts the estimate by a between-cell covariance term. On 229,605 Visual Genome relations a class-only network improves the frequency baseline entirely through the prior channel, while a frozen-CLIP model scoring 0.04655 *worse* overall has significantly *better* instance alignment (+0.00641, $p = .002$) than the box-geometry model it loses to. Applied unchanged to long-tailed classification, it separates two re-weighting schedules that aggregate scores cannot: one whose near-zero total gain conceals a +0.19713 prior gain cancelling a -0.19569 alignment loss, and one whose genuine improvement is two-thirds prior-recoverable and one-third instance alignment.

Keywords: Visual Reasoning, Dataset Bias, Counterfactual Evaluation, U-statistics, Randomization Tests, Scene Graph Generation

1. Introduction

Computer-vision benchmarks often contain a shortcut that is both powerful and easy to state. In scene graph generation (SGG), the ordered object-class pair strongly predicts the relation label; in visual question answering (VQA), question form predicts the answer; in long-tail recognition, category frequency dominates aggregate performance. Models can therefore improve a headline score by fitting annotation regularities more precisely without improving the instance-specific association between an image and its label. This is not merely hypothetical: frequency baselines are competitive on Visual Genome [43], VQA systems fail under changed answer priors [16, 1], standard

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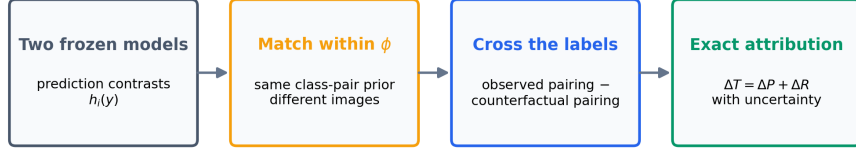


Figure 1: **WAGER audits the gain directly.** Prediction contrasts from two frozen models are matched within a declared prior cell, evaluated under observed and crossed label assignments, and separated exactly into prior-transported and instance-alignment components.

metrics obscure tail behavior [27], and the same diagnosis continues to reappear in current work: open-vocabulary and knowledge-enhanced SGG methods still report a single aggregate score [26, 44, 34], long-tailed recognition methods rebalance training toward rare classes without attributing a resulting gain to the frequency prior versus instance-specific evidence [29, 25, 28], and recent benchmark audits show multimodal models can exploit non-visual shortcuts that inflate exactly this kind of aggregate score [5, 45].

The usual audit trains or constructs a prior-only baseline and compares its score with a full model. That strategy is useful diagnostically, but it does not identify where the *gain between two submitted systems* came from. Baseline estimation error, smoothing, calibration, and model-family mismatch all enter the result. More fundamentally, separately decomposing each model and subtracting the decompositions is an indirect answer to a pairwise question.

We instead ask a counterfactual question at the level of the gain itself. Let f_0 be the old model, f_1 the claimed improvement, and $\phi(x)$ the declared prior feature. For two test examples i, j with the same ϕ , we retain both models’ prediction vectors but cross the observed labels. The crossed assignment preserves the prior cell and its label histogram while destroying which prediction vector belongs with which label. Comparing the observed and crossed assignments measures whether $f_1 - f_0$ improves this instance-level alignment. Figure 1 summarizes the idea.

This construction gives WAGER: *Within-cell Antisymmetric Gain Evaluation of Reasoning*. “Reasoning” is used operationally: predictive alignment with the correct instance beyond ϕ , not a claim of human-like cognition or causality. WAGER makes five contributions.

1. We define a direct model-pair estimand whose prior component is generated by within-cell label transport, not by fitting a frequency forecaster.
2. We prove an exact population and finite-sample decomposition. The residual is an antisymmetric U-statistic; under the quadratic score it has a covariance identity that identifies it with the resolution term of the classical proper-score decomposition [30, 11, 4], applied to a model-pair contrast rather than to one forecaster.
3. We prove two further exact results that were not available for the single-model

decomposition: leave-one-out transport removes a finite-sample attenuation bias of $(n_c - 1)/n_c$ that a naive in-sample plug-in prior would otherwise incur at every finite cell size (Proposition 2), and coarsening the audit cell changes the estimate by an explicit between-cell covariance term (Proposition 1), which turns an empirical sensitivity finding into a population-level mechanism.

4. We derive an $O(NK)$ algorithm, a Hájek influence function for image-cluster-robust intervals, and a cell-stratified randomization test under a sharp exchangeability null.
5. We validate the estimator under controlled alternatives and on Visual Genome, where it distinguishes a class-only gain from gains created by adding spatial evidence, and further from gains created by a model that consumes real image pixels through a frozen CLIP encoder rather than only box geometry (Section 4.5).
6. We show WAGER is not SGG-specific by applying it unchanged to a second domain and task type – long-tailed image classification on CIFAR-100-LT – decomposing the gain of a class-balanced classifier over a vanilla cross-entropy baseline into class-frequency-prior and instance-alignment components (Section 4.6), giving empirical support to the broader-applicability claim of Section 5.2 rather than leaving it as an untested hypothesis.

The redesign is important in what it removes. WAGER does not estimate a model’s “self-prior,” does not require a projection/evaluation split, does not equate betting growth with information, and does not rely on an arbitrary threshold for a gain ratio. The primary quantity is the signed instance-alignment gain with uncertainty.

2. Related Work

2.1. Benchmark priors and shortcut learning

Dataset bias has long complicated comparisons between visual recognition systems [38, 14]. In SGG, the FREQ predictor estimates $p(\text{predicate} \mid \text{subject, object})$ and is surprisingly competitive [43]; mean recall and causal interventions were proposed to reduce this dependence [9, 36, 35], although metric pathologies remain [27]. More recent SGG work continues this line under open-vocabulary and knowledge-enhanced settings [26, 44, 34], and still reports gains as a single aggregate score rather than attributing them to prior fitting versus instance-specific evidence. VQA-CP changes language priors between train and test [1], but changing-prior splits can themselves be exploited [37]; recent VQA work continues to target the same language-prior shortcut through contrastive and relation-aware objectives [41, 24]. Benchmark-level audits of exploitable non-visual shortcuts have also expanded beyond VQA to broader multi-modal evaluation [5, 45], reinforcing the same diagnosis WAGER targets: an aggregate score can rise for reasons unrelated to instance-specific visual evidence. These methods alter training, data, or metrics, or audit a single model’s exploitability against a benchmark’s own structure. WAGER instead audits a fixed pair of probabilistic predictors on a fixed test set, attributing their score difference rather than characterizing either model alone.

2.2. Long-tailed recognition

The same shortcut, in a structurally identical form, appears well outside scene graph generation. A long-tailed classifier can raise its aggregate accuracy simply by fitting the training-time class-frequency prior more precisely, without becoming any better at discriminating instances within a class [29, 12], and the remedies proposed for this – class-balanced re-weighting, logit adjustment, feature fusion – all work by rebalancing the training objective or the learned representation toward rare classes [29, 25, 28], much as mean-recall and causal-intervention methods rebalance SGG training. What such remedies share with their SGG counterparts is that they intervene on how a model is trained rather than on how a trained pair is compared, so none of them attributes an observed accuracy gap between two finished models to frequency-fitting on the one hand and instance-specific improvement on the other. That is the question Section 4.6 puts to WAGER, using the benchmark’s own coarse superclasses as the declared prior feature ϕ ; the estimator turns out to separate two re-weighting schedules whose aggregate scores would not distinguish them at all, one whose near-zero total gain conceals a large prior improvement cancelling an equally large alignment loss, the other whose genuine improvement is part prior-refitting and part rare-class alignment.

2.3. Information and predictive probing

Usable-information analyses measure information accessible to a chosen predictor family [42, 13, 19]. Their central objects are model- or family-level quantities. WAGER is deliberately narrower and more operational: it attributes the proper-score difference between two concrete frozen models. We do not identify conditional mutual information and do not claim that a positive residual establishes causal or compositional reasoning.

2.4. Proper scores, randomization, and U-statistics

Strictly proper scoring rules reward honest predictive distributions [15]; WAGER uses their model-pair contrast. A recent survey revisits estimation and forecast evaluation under proper scoring rules broadly [40], and the reliability–resolution–uncertainty partition itself continues to be re-derived and extended: to general strictly proper scores and multiclass outcomes [8], to entropy/divergence-based uncertainty quantification [21], and via an alternative covariance rearrangement for the Brier score specifically [39]. WAGER uses the same partition but applies it to a paired gain contrast rather than to one forecaster, as detailed in Section 2.5. Permutation methods have been used to assess classifier performance and conditional independence [31, 3], with recent work characterizing when generalized permutation tests attain optimal power [22] and extending conditional-independence testing via ball-divergence measures [2] and transport-map reductions to the unconditional case [17]. Conditional permutation tests typically estimate or assume a conditional distribution when confounders are continuous or high-dimensional. WAGER exploits a different setting: the benchmark supplies a discrete audit cell ϕ , and the target is not generic conditional independence but a specific model-to-model score gain. This permits exact within-cell transport without estimating a nuisance law. The antisymmetric residual is an order-two U-statistic in the sense of Hoeffding [20]; its Hájek-projection inference is part of the

same broader program of finite- and large-sample U-statistic theory that recent work extends to anytime-valid confidence sequences [6], enabling an efficient all-pairs estimator and influence-function inference here. Pairwise betting has also been used to test exchangeability sequentially [33]; WAGER is a fixed-sample gain decomposition and makes no sequential-validity claim. Its new element is the gain-contrast transport identity and the resulting attribution algorithm, rather than permutation or U-statistic theory by itself.

2.5. Relation to classical proper-score decompositions

The covariance identity in Eq. (7) is a relative of the reliability resolution-uncertainty partition of a single proper score [30, 11], later generalized and made conditioning-set explicit by Bröcker [4]: conditioned on a chosen grouping variable, a proper score splits into a calibration term and a resolution term, and the resolution term is a covariance between the forecast and the outcome indicator within groups. We deliberately build on this fact rather than obscure it. WAGER is not a new proper score, and ΔR_c is best understood as a *resolution difference*, not a new quantity to be evaluated on its own. Two things distinguish WAGER from applying the classical decomposition to q_0 and q_1 separately and subtracting the two resolutions. First, that indirect route requires two resolution estimates, each of which is generally biased in finite samples by the same self-prediction mechanism formalized in Section 3.6; the bias does not cancel under subtraction because it scales with each model’s own cell counts. WAGER instead forms the paired gain vector $H_x(y)$ first and decomposes the *contrast* in one pass, which is what makes the exact finite-sample identity and the antisymmetric U-statistic kernel possible. Second, the classical decomposition is a diagnostic for a single forecaster’s calibration and sharpness; it neither targets nor equips a specific pairwise comparison with inference. WAGER supplies image-cluster-robust confidence intervals and a cell-stratified randomization test for exactly the gain a benchmark submission claims. The technical contribution here is narrower than a new form of score decomposition: it is the transport construction and finite-sample identity that convert a known resolution concept into a model-to-model attribution algorithm with an exactness and inference guarantee that the single-model decomposition does not provide.

3. Method

3.1. Setup: audit a gain, not a model

Let $q_0(\cdot | x)$ and $q_1(\cdot | x)$ be the predictive distributions of an old and a new frozen model over K labels. Let $S(q, y)$ be a bounded proper score, oriented so that larger is better, and let $C = \phi(X)$ be a declared discrete prior feature. For SGG, C is the ordered subject-object class pair. Define the instance-specific *gain vector*

$$H_x(y) = S(q_1(\cdot | x), y) - S(q_0(\cdot | x), y), \quad y = 1, \dots, K. \quad (1)$$

Unlike the realized gain $H_X(Y)$, this vector can be evaluated at a counterfactual label without rerunning either model.

Our default is the quadratic score

$$S_B(q, y) = 2q_y - \|q\|_2^2, \quad (2)$$

which differs from negative Brier loss only by a label-only constant. It is strictly proper and bounded, avoids probability clipping, and uses the entire predictive distribution. A floored log score is a sensitivity analysis.

3.2. Within-cell counterfactual transport

For a prior cell c , define three population quantities:

$$\Delta T_c = \mathbb{E}[H_X(Y) \mid C = c], \quad (3)$$

$$\Delta P_c = \mathbb{E}_{H|c} \mathbb{E}_{Y'|c}[H_X(Y')], \quad (4)$$

$$\Delta R_c = \Delta T_c - \Delta P_c. \quad (5)$$

Here Y' is an independent label drawn from the same cell. Thus ΔT_c is the observed score gain, ΔP_c is the gain retained after breaking the instance-specific prediction–label assignment while preserving the prior cell, and ΔR_c is the alignment gain lost by that transport.

Theorem 1 (Transport and covariance identities). *For any integrable score contrast H ,*

$$\Delta T_c = \Delta P_c + \Delta R_c, \quad \Delta R_c = \sum_{y=1}^K \text{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C = c). \quad (6)$$

For the quadratic score, writing $\Delta q_y(X) = q_1(y \mid X) - q_0(y \mid X)$,

$$\Delta R_c = 2 \sum_{y=1}^K \text{Cov}(\Delta q_y(X), \mathbb{1}\{Y = y\} \mid C = c). \quad (7)$$

Consequently, if the gain vector is a function of c alone, then $\Delta R_c = 0$.

Equation (7) is the operational meaning of WAGER: the new model receives reasoning credit only to the extent that its probability change aligns more strongly with the label of the correct instance within the same prior cell. Cell-constant score improvements, including improved fitting of the cell’s label histogram, remain in ΔP . The result does not assert that every nonzero covariance is causal reasoning; unrecorded within-cell shortcuts can also contribute, so ϕ must be declared and stress-tested.

3.3. Coarsening the audit cell

Section 4.4 reports that replacing the class-pair cell with a coarser subject-only cell changes the estimated alignment gain. The following result gives that sensitivity an exact population-level mechanism rather than treating it as an empirical curiosity.

Proposition 1 (Coarsening decomposition). *Let $\bar{\phi} = g(\phi)$ be any coarsening of the audit feature, so that each cell $\bar{c}$ of $\bar{\phi}$ is a union of cells of ϕ . Then*

$$\Delta R_{\bar{c}} = \mathbb{E}[\Delta R_c \mid \bar{\phi} = \bar{c}] + \sum_{y=1}^K \text{Cov}(\mathbb{E}[H_X(y) \mid C], \text{Pr}(Y = y \mid C) \mid \bar{\phi} = \bar{c}). \quad (8)$$

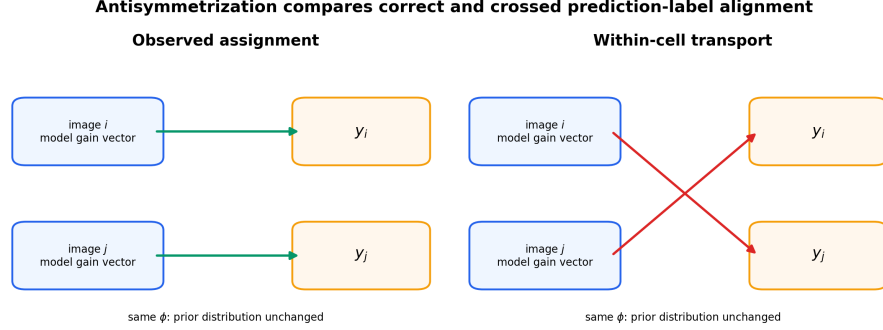


Figure 2: **The antisymmetric pair kernel.** For two examples in the same prior cell, WAGER subtracts the score under crossed labels from the score under the observed assignment. Cell membership and label counts are unchanged; only instance alignment is removed.

The first term is the coarse cell’s share of the fine-grained alignment gain the paper already reports; it is the quantity a reader would expect a coarser audit to recover on average. The second term is new: it is a between-cell covariance, taken over the finer cells nested inside $\bar{c}$, between each fine cell’s typical gain at label y and its label- y frequency. It is generally not sign definite, so coarsening ϕ can inflate or deflate the credited alignment gain relative to the finer audit, and Table 3 shows the empirically inflating case. Because this second term vanishes identically once ϕ is refined to make it constant on every non-vanishing fine cell (in particular, in the limit where ϕ is the finest cell that remains non-singleton for every example), Proposition 1 formalizes why the finest defensible audit feature is the conservative default: coarsening can only add cross-cell prior structure into ΔR , never remove genuine within-cell alignment that the fine partition already isolates. The proof is a per-label application of the law of total covariance, summed over y (Appendix A).

3.4. Exact finite-sample decomposition

Consider cell c with $n_c \geq 2$ examples. Write $H_i(y) = H_{x_i}(y)$. WAGER computes

$$\widehat{\Delta T}_c = \frac{1}{n_c} \sum_{i \in c} H_i(y_i), \quad (9)$$

$$\widehat{\Delta P}_c = \frac{1}{n_c(n_c - 1)} \sum_{i \neq j \in c} H_i(y_j), \quad (10)$$

$$\widehat{\Delta R}_c = \binom{n_c}{2}^{-1} \sum_{i < j \in c} A_{ij}, \quad (11)$$

where the antisymmetric kernel is

$$A_{ij} = \frac{1}{2} \{H_i(y_i) + H_j(y_j) - H_i(y_j) - H_j(y_i)\}. \quad (12)$$

Theorem 2 (Finite-sample WAGER identity). *For every cell with $n_c \geq 2$, without a distributional assumption,*

$$\widehat{\Delta T}_c = \widehat{\Delta P}_c + \widehat{\Delta R}_c. \quad (13)$$

Moreover, under exchangeable sampling within the cell, $\widehat{\Delta R}_c$ is an unbiased order-two U-statistic for ΔR_c .

The dataset-level quantities weight cells by their number of identified examples: $\widehat{\Delta Z} = N_*^{-1} \sum_{c: n_c \geq 2} n_c \widehat{\Delta Z}_c$ for $Z \in \{T, P, R\}$. Singleton cells cannot identify a within-cell counterfactual; WAGER excludes and reports them instead of inventing a smoothed estimate. If $\widehat{\Delta T} > 0$, the descriptive alignment share is $\widehat{\rho} = \widehat{\Delta R} / \widehat{\Delta T}$. It may exceed one when instance alignment improves while prior fitting deteriorates, and it is undefined for a nonpositive total gain. We therefore report $\widehat{\Delta R}$ as the primary result, not a thresholded ratio.

3.5. Linear-time computation

Naively evaluating Eq. (10) is quadratic in cell size. Let m_{cy} be the count of label y in cell c . For example $i \in c$, its transported score is

$$P_i = \frac{\sum_{y=1}^K m_{cy} H_i(y) - H_i(y_i)}{n_c - 1}. \quad (14)$$

Averaging P_i yields Eq. (10); averaging $H_i(y_i) - P_i$ yields Eq. (11). Thus the entire estimator costs $O(NK)$ time and $O(NK)$ storage for cached prediction vectors. No model fitting, projection fold, hierarchy, or smoothing hyperparameter appears.

3.6. Why leave-one-out transport, not a fitted-in-sample prior

Equation (14) excludes example i 's own label from its transported score. It is natural to ask what is lost by the simpler alternative of fitting the cell's empirical label frequency on all n_c examples, including i , and using that fitted frequency as the counterfactual, i.e. treating $\hat{p}_c(y) = m_{cy}/n_c$ as a prior model and setting $\tilde{P}_i = \sum_y \hat{p}_c(y) H_i(y) = n_c^{-1} \sum_y m_{cy} H_i(y)$. This is the natural plug-in estimator, and it is what a frequency-collapsed projection baseline computes when no separate evaluation fold is withheld from fitting.

Proposition 2 (Exact attenuation of the in-sample plug-in). *For every cell with $n_c \geq 2$ and every $i \in c$, writing $\hat{P}_i$ for Eq. (14) and $\hat{R}_i = H_i(y_i) - \hat{P}_i$,*

$$\tilde{P}_i = \hat{P}_i + \frac{1}{n_c} \hat{R}_i, \quad \tilde{R}_i := H_i(y_i) - \tilde{P}_i = \frac{n_c - 1}{n_c} \hat{R}_i. \quad (15)$$

Consequently the in-sample plug-in's dataset-level statistic obeys $\widetilde{\Delta R} = N_^{-1} \sum_{c: n_c \geq 2} (n_c - 1) \widehat{\Delta R}_c$, a multiplicative shrinkage of every cell's leave-one-out alignment gain by the factor $(n_c - 1)/n_c$.*

The shrinkage in Eq. (15) is deterministic, not a variance statement: whenever $\widehat{\Delta R}_c \neq 0$, the in-sample plug-in is biased for it at every finite n_c , with the largest relative attenuation exactly in the smallest identified cells (50% at $n_c = 2$, vanishing only as $n_c \rightarrow \infty$). This is why the redesign removes the projection/evaluation split that the rejected precursor of WAGER used to control the same self-prediction bias: a held-out fold restores unbiasedness by fitting $\hat{p}_c$ on data disjoint from the scored examples, but it does so by discarding examples from both stages and by introducing a split-fraction hyperparameter and, typically, higher variance. WAGER’s leave-one-out U-statistic removes the bias exactly, at the full sample, for every ϕ -cell with $n_c \geq 2$, without a fitted nuisance distribution at all. Section 2.5 situates this result against the classical single-model score decomposition, where the analogous self-prediction bias affects each model’s resolution estimate separately.

3.7. Inference

For point estimation we use all unordered within-cell pairs. For uncertainty, define $\bar{A}_i = (n_c - 1)^{-1} \sum_{j \neq i} A_{ij}$ and $\widehat{\Delta R}_c = n_c^{-1} \sum_i \bar{A}_i$. The estimated Hájek influence contribution for random cell membership is

$$\widehat{\psi}_i = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}. \quad (16)$$

We sum $\widehat{\psi}_i$ within image and use the resulting one-way sandwich variance, so multiple relations from the same image are not treated as independent. Total-gain influence is $H_i(y_i) - \widehat{\Delta T}$; prior-gain and share intervals follow from the exact identity and delta method.

As a complementary finite-group diagnostic, WAGER applies an independently sampled cyclic shift to labels inside every cell. Each shift preserves cell size and label histogram. Recomputing $\widehat{\Delta R}$ under B shifts gives

$$p = \frac{1 + \sum_{b=1}^B \mathbb{1}\{\widehat{\Delta R}^{(b)} \geq \widehat{\Delta R}\}}{B + 1}. \quad (17)$$

This is exact under the sharp null that labels are exchangeable relative to the frozen model outputs within each cell. Because SGG relations share images, our primary real-data uncertainty statement is the image-cluster-robust interval; the randomization test is reported as a secondary relation-level diagnostic.

4. Experiments

We evaluate five questions: (i) does WAGER remain centered under the null and respond monotonically to known instance signal; (ii) does it distinguish class-prior fitting from spatial evidence on a real long-tailed benchmark; (iii) are conclusions stable to score, cell granularity, and sparse-cell filtering; (iv) does the alignment channel behave sensibly when a model consumes real image pixels rather than annotation-derived geometry; and (v) does the same estimator, unchanged, produce a meaningful attribution on a structurally different task in another domain? Questions (i)–(iii) use Visual Genome and controlled simulations; (iv) adds a frozen CLIP encoder on real VG images; (v) moves to long-tailed image classification on CIFAR-100-LT. All confidence intervals are 95%.

Algorithm 1 WAGER gain decomposition

Require: frozen q_1, q_0 ; labels y ; prior cells ϕ ; proper score S

Ensure: $\widehat{\Delta T}, \widehat{\Delta P}, \widehat{\Delta R}$ and uncertainty

- 1: $H_i(k) \leftarrow S(q_{1i}, k) - S(q_{0i}, k)$ for all i, k
 - 2: **for** each cell c with $n_c \geq 2$ **do**
 - 3: count labels $m_{c1}, \dots, m_{cK}$
 - 4: **for** each $i \in c$ **do**
 - 5: $P_i \leftarrow (\sum_k m_{ck} H_i(k) - H_i(y_i)) / (n_c - 1)$
 - 6: $R_i \leftarrow H_i(y_i) - P_i$
 - 7: **end for**
 - 8: **end for**
 - 9: $\widehat{\Delta T} \leftarrow \text{mean } H_i(y_i)$; $\widehat{\Delta P} \leftarrow \text{mean } P_i$; $\widehat{\Delta R} \leftarrow \text{mean } R_i$
 - 10: compute image-clustered Hájek interval and cell-shift randomization p
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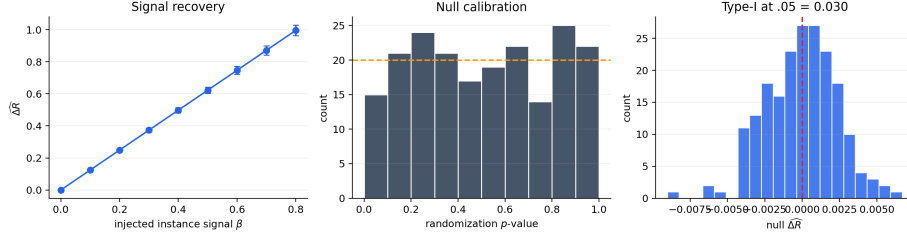


Figure 3: **Controlled validation.** WAGER recovers injected within-cell signal monotonically (mean $\pm$ one standard deviation), produces approximately uniform null randomization p -values, and controls type-I error in 200 null runs.

4.1. Controlled validation

We generate 24 prior cells over five labels, drawing each cell’s label distribution from a Dirichlet law and taking the old model to be exactly the cell prior. The new model interpolates toward an oracle, $q_1 = (1 - \beta)q_0 + \beta q_{\text{oracle}}$ with q_{oracle} placing 0.98 on the realized label, and forty repetitions are run at each $\beta \in \{0, .1, \dots, .8\}$. Across that range the estimated alignment gain grows linearly and monotonically with the injected signal (Figure 3, left).

Type-I behaviour is assessed by perturbing the prior predictions with mean-zero noise and then randomly reassigning the perturbed vectors within each cell, which guarantees that no systematic prediction–label alignment survives. Over 200 such datasets the 99-shift randomization test rejects in 0.030 of runs at level 0.05, with a mean estimated alignment gain of -8.97×10^{-5} ; the null p -values come out approximately uniform and the null estimates stay centred at zero (Figure 3, centre and right).

4.2. Visual Genome setup

We work with Visual Genome [23] in the standard VG150 PredCls regime, where ground-truth boxes and object labels are supplied and the model has only to predict one of 50 predicate labels for each ordered object pair. The split provides 532,987 training

Table 1: Frozen-model top-1 accuracy on VG150 PredCls. Similar accuracies do not reveal whether the difference is prior-recoverable.

Model	Accuracy
FREQ	0.6564
FREQ+OVERLAP	0.6545
MLP-CLASS	0.6554
MLP-SPATIAL	0.6639
MLP-SPATIAL+	0.6674

and 229,605 test relations, drawn from 65,228 and 27,955 images respectively, and the prior cell throughout is $\phi = (\text{subject class}, \text{object class})$, two real examples of which appear in Figure 4. Of the 8,614 cells present in the test split, 6,346 contain at least two relations and are therefore identified; the remaining 2,268 are singletons, which WAGER excludes and reports rather than smoothing over, leaving 227,337 relations (99.0%) under audit.

The five predictors are deliberately constructed so that each exposes a different kind of information. FREQ is simply the training-count distribution conditioned on the class pair, and FREQ+OVERLAP adds to it a single binary box-overlap indicator. MLP-CLASS consumes fixed subject and object class embeddings, which makes it a function of ϕ alone; MLP-SPATIAL extends those embeddings with 12 normalized box-geometry features covering relative centre, scale, aspect, intersection, containment and IoU, while MLP-SPATIAL+ merely increases capacity. Each is trained once on the training split, and WAGER thereafter consumes nothing but their cached test distributions, scored quadratically with image-clustered intervals and 499 within-cell cyclic randomizations.

4.3. Direct gain attribution

The central result is collected in Table 2 and Figure 5. MLP-CLASS improves on FREQ by 0.03386 of quadratic score, yet because its gain vector is constant inside every class-pair cell, the estimator assigns that entire improvement to prior transport and precisely nothing to instance alignment. The zero is worth dwelling on: it follows algebraically from the construction rather than from a learned baseline happening to pass a sanity check.

Adding box geometry changes the picture. MLP-SPATIAL gains 0.04270 over FREQ, of which 0.02832 survives label transport and 0.01439 is instance alignment (CI [0.01373, 0.01504]). More revealing still is the direct comparison against MLP-CLASS, where the total gain falls to 0.00885 only because prior fitting deteriorates by 0.00554 even as spatial alignment improves by 0.01439. The resulting alignment share of 1.63 records a genuinely useful instance-specific change that the aggregate score partly conceals behind a worse prior component, and it is exactly the configuration that a ratio confined to $[0, 1]$, or thresholded at 0.5, would misdescribe.

The remaining two comparisons bracket the interesting range. MLP-SPATIAL+ adds 0.00374 over MLP-SPATIAL, nearly all of it alignment at 0.00321 (CI [0.00289, 0.00353]), whereas FREQ+OVERLAP supplies the opposite caution: its

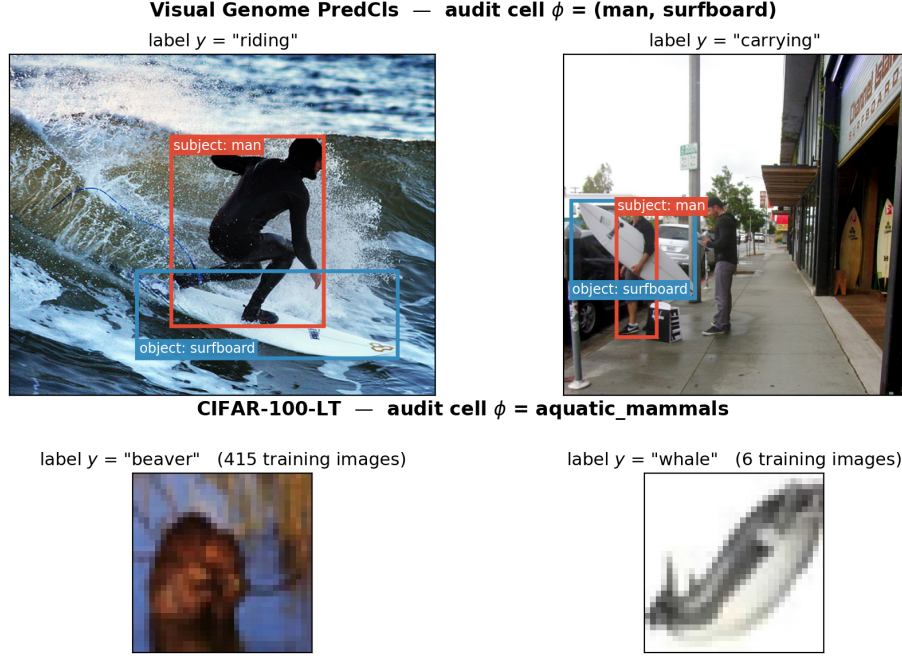


Figure 4: **What an audit cell contains, on both benchmarks.** Each row shows two real test examples that share the same declared prior feature ϕ but carry different labels – precisely the configuration within-cell transport exploits. *Top*: two Visual Genome relations from the cell $\phi = (\text{man}, \text{surfboard})$. The class pair is identical and therefore cannot distinguish “riding” from “carrying”; only the image content can, which is what ΔR is designed to credit. Boxes are the annotated subject (red) and object (blue) regions that MLP-SPATIAL turns into geometry features and MLP-VISUAL crops and encodes with CLIP. *Bottom*: two CIFAR-100 test images from the superclass $\phi = \text{aquatic_mammals}$, drawn from opposite ends of the long-tailed training distribution (415 versus 6 training images). Crossing the labels within a row preserves the cell and its label histogram while destroying which prediction belongs with which instance.

overlap indicator does create a small positive alignment term, but the accompanying negative prior term is larger, so the model ends up worse overall. Using image-derived information is thus never treated as an improvement in itself; both channels are reported, signs included.

4.4. Sensitivity analyses

For MLP-SPATIAL versus MLP-CLASS, the conclusion survives every planned sensitivity check (Table 3). Changing from the quadratic to a probability-floored log score changes scale (0.01439 to 0.06536) but not sign or significance. Coarsening ϕ from the class pair to subject class increases the credited alignment to 0.02181. Proposition 1 identifies why: the coarser cell’s estimate equals the class-pair estimate’s conditional average plus a between-class-pair covariance term, and that term is positive here, meaning some of the credited gain reflects class-pair-level structure that the finer audit had already correctly assigned to ΔP . This is precisely the sense in which the class-pair audit is the conservative choice: it is not that coarsening is wrong, but that it cannot

Table 2: WAGER decomposition on 227,337 identified VG150 PredCls relations. Gains use the quadratic score. CI is image-cluster robust; p is the 499-shift relation-level randomization diagnostic. “–” denotes an undefined share because total gain is negative.

New model	Old model	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	$\widehat{\rho}$	p
FREQ+OVERLAP	FREQ	−0.01229	−0.01430	0.00202 [0.00168, 0.00235]	–	.002
MLP-CLASS	FREQ	0.03386	0.03386	0.00000 [0.00000, 0.00000]	0.00	.724
MLP-SPATIAL	FREQ	0.04270	0.02832	0.01439 [0.01373, 0.01504]	0.34	.002
MLP-SPATIAL+	FREQ	0.04644	0.02885	0.01760 [0.01686, 0.01834]	0.38	.002
MLP-SPATIAL	MLP-CLASS	0.00885	−0.00554	0.01439 [0.01373, 0.01504]	1.63	.002
MLP-SPATIAL+	MLP-SPATIAL	0.00374	0.00053	0.00321 [0.00289, 0.00353]	0.86	.002

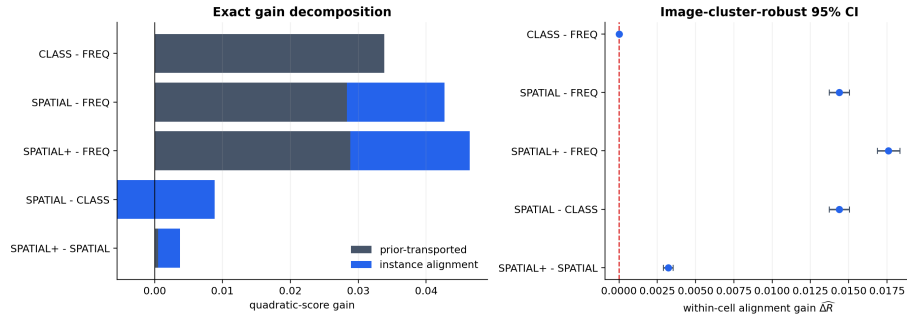


Figure 5: **Direct model-pair attribution.** Left: exact decomposition of each positive gain into prior-transported and instance-alignment components (the negative prior term for SPATIAL–CLASS extends left of zero). Right: primary alignment estimates with image-cluster-robust intervals.

be assumed neutral, and only the finer partition’s estimate isolates alignment gain net of every prior regularity expressible in ϕ . Raising the minimum cell size from 2 to 5, 10, and 20 yields 0.01395, 0.01330, and 0.01225; all cluster-robust lower limits remain positive. Unlike the rejected projection-based formulation, there is no smoothing strength, split fraction, clipping constant, betting order, or gain-ratio decision threshold to tune.

4.5. Real-pixel signal on Visual Genome

None of the models in Sections 4.2–4.4 ever sees an image. They are functions of the declared class pair and, in the case of MLP-SPATIAL(+), of box coordinates recorded in the annotation. Since box geometry serves only as a proxy for what an image contains, a sceptical reader is entitled to ask whether the alignment channel is responding to genuine pixel evidence or merely to that proxy. Figure 4 sharpens the question: the cell $\phi = (\text{man}, \text{surfboard})$ holds both “riding” and “carrying”, and although their box configurations certainly differ, what actually distinguishes the two relations is what the photograph shows.

MLP-VISUAL is introduced to answer it. Subject and object regions are cropped from the original Visual Genome image and each is encoded by a frozen CLIP ViT-B/32 image encoder [32], after which the two 512-dimensional embeddings are concatenated with the class embeddings that MLP-CLASS already uses and passed to a small trained

Table 3: Sensitivity of the MLP-SPATIAL vs. MLP-CLASS decomposition to score, audit granularity, and minimum cell size. CI is image-cluster robust on ΔR .

Factor	Setting	$\widehat{\Delta T}$	$\widehat{\Delta R}$ (95% CI)	Cells
Score	Brier (default)	0.00885	0.01439 [0.01373, 0.01504]	6346
Score	Log (floored)	0.04297	0.06536 [0.06294, 0.06779]	6346
Audit cell	class pair (default)	0.00885	0.01439 [0.01373, 0.01504]	6346
Audit cell	subject only	0.00943	0.02181 [0.02079, 0.02283]	150
Min. cell size	2 (default)	0.00885	0.01439 [0.01373, 0.01504]	6346
Min. cell size	5	0.00772	0.01395 [0.01329, 0.01462]	4060
Min. cell size	10	0.00683	0.01330 [0.01262, 0.01398]	2744
Min. cell size	20	0.00570	0.01225 [0.01155, 0.01295]	1757

head of 256 and 128 units. Because CLIP itself is never fine-tuned, whatever alignment the model gains reflects information its pretraining had already recovered from pixels and the head has merely matched to an SGG label, rather than representation learning performed on Visual Genome.

Matched training data.. We report this comparison apart from Table 2 rather than as an additional row, for a reason that is itself an instance of the paper’s argument. Encoding every training relation with CLIP is expensive enough that MLP-VISUAL must be trained on a subsample, and setting a subsampled model against the full-data MLP-SPATIAL would confound feature content with training-set size so thoroughly that a disappointing result could not honestly be attributed to the pixels. All three compared predictors are therefore retrained on a single fixed, seeded subsample of 100,000 of the 532,987 training relations, and are written MLP-CLASS-S, MLP-SPATIAL-S and MLP-VISUAL-S to mark the fact. Differing only in their feature sets, any gap between them isolates the features. The audit continues to use the complete 229,605-relation test split with the same class-pair ϕ and image-clustered intervals, so its numbers remain directly comparable with the main analysis. The sampling procedure, the crop-deduplication scheme and the count of images that could not be retrieved are given in Appendix Appendix D.

Result.. Judged by any aggregate measure, the CLIP model is the worst of the three. Its top-1 accuracy is 0.6161 against 0.6467 for MLP-SPATIAL-S and 0.6416 for MLP-CLASS-S, and its quadratic score is lower still, falling 0.04655 below MLP-SPATIAL-S. An evaluator working from either number would conclude that replacing box geometry with pixel content had simply made the predictor worse, and would discard it.

The decomposition in Table 4 and Figure 6 contradicts that reading on the axis the paper cares about. Against MLP-SPATIAL-S the CLIP model loses 0.05296 of prior-transported gain while *gaining* 0.00641 of instance alignment (95% CI [0.00514, 0.00769], randomization $p = .002$), so its entire deficit sits in the prior channel and its instance-level discrimination is significantly better, not worse. The comparison against the class-only baseline tells the same story more plainly: pixels

Table 4: Matched-subsample real-pixel study on the full VG150 test split (coverage 99.0%, as in Table 2). All three predictors are trained on the same 100,000 relations and differ only in features. CI is image-cluster robust on $\widehat{\Delta R}$; p is the 499-shift randomization diagnostic.

New model	Old model	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
MLP-SPATIAL-S	MLP-CLASS-S	0.00503	-0.00520	0.01023 [0.00967, 0.01079]	.002
MLP-VISUAL-S	MLP-CLASS-S	-0.04152	-0.05816	0.01665 [0.01534, 0.01796]	.002
MLP-VISUAL-S	MLP-SPATIAL-S	-0.04655	-0.05296	0.00641 [0.00514, 0.00769]	.002

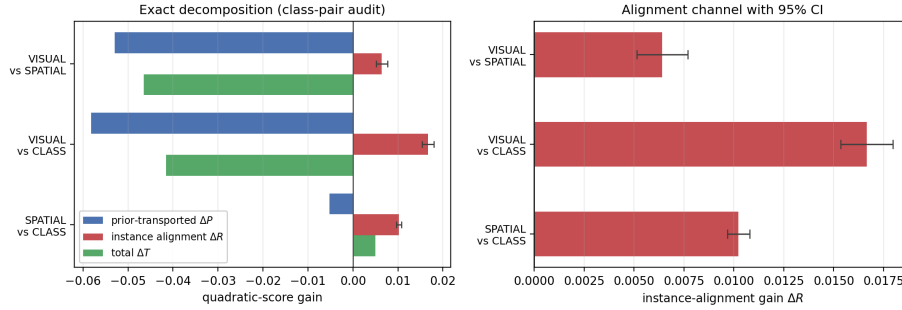


Figure 6: **Pixels beat geometry on alignment while losing on the aggregate.** Left: the exact decomposition for the three matched-subsample pairs. Both comparisons involving the CLIP model have strongly negative totals (green) driven entirely by the prior channel (blue), while their alignment channel (red) is positive in every case. Right: the alignment channel alone, with 95% image-clustered intervals. Replacing box geometry with real image content raises instance alignment from 0.01023 to 0.01665 over the same class-only baseline.

buy 0.01665 of alignment where box geometry buys 0.01023, and the two intervals are far apart. Real image content therefore carries more instance-specific signal than the geometric proxy standing in for it, which is precisely what Section 4.5 set out to test.

Why the prior channel collapses is not mysterious. The 1024 CLIP dimensions dominate the 64 dimensions of class embedding that the head also receives, so the model fits the class-pair frequency structure of VG150 markedly less well than a predictor built around that structure. A benchmark whose headline metric rewards prior fitting will punish this model, and the aggregate numbers duly do. What WAGER adds is the observation that the punishment is not for failing to use the image; it is for failing to reproduce the annotation prior, while using the image better than either competitor.

This is the paper’s argument running in the opposite direction from Section 4.6. There a near-zero total gain concealed a large alignment *loss*; here a clearly negative total gain conceals a real alignment *gain*. In both cases the aggregate score is not merely imprecise but actively misleading about the thing a vision benchmark is meant to measure, and in both cases the two channels recover what it hides.

4.6. Cross-domain validation: long-tailed image classification

Sections 4.2–4.5 are all instances of one task (predicate classification on an ordered object pair). Section 2.2 argued that long-tailed image classification poses a structurally

identical question: a class-frequency prior can be fit better without any per-instance improvement. We test this directly on CIFAR-100-LT [10], an exponential-imbalance (ratio 100) subsample of CIFAR-100’s training split with the original, balanced test split retained for evaluation. We train ResNet-32 [18] classifiers that differ only in their per-class loss weighting: a vanilla cross-entropy baseline (CE); class-balanced effective-number re-weighting applied from initialization (CB) [10]; and the same re-weighting deferred to the final twenty epochs (DRW) [7]. Optimizer, schedule, augmentation, epoch budget, and seed are identical across runs (Appendix Appendix D).

Choosing ϕ . A declared prior feature has to predict the label without determining it, and this second benchmark makes the constraint unusually easy to violate. Were ϕ taken to be the fine class label itself, every audit cell would be label-homogeneous, within-cell transport would have nothing to cross, and $\widehat{\Delta R}$ would come out identically zero – an algebraic artifact wearing the appearance of a finding. CIFAR-100’s own 20 coarse superclasses avoid this, each gathering five fine labels into a cell of 500 balanced test images, which is structurally the situation of a VG class-pair cell holding several predicates (Figure 4). Alongside the superclass audit we report two further declared priors: the trivial coarsening to a single global cell, which audits against the global class marginal and so instantiates Proposition 1 in a second domain, and a partition into head, medium and tail training-frequency tiers. CIFAR-100 test images are independent, unlike the several relations VG draws from one photograph, so no cluster grouping is required and the plain Hájek interval of Section 3.7 applies directly.

Result. The three classifiers reach top-1 accuracies of 0.3662 (CE), 0.2627 (CB) and 0.3874 (DRW), so re-weighting from initialization degrades the classifier substantially while the deferred schedule delivers the improvement it was designed to produce. Table 5 decomposes both gains against the superclass audit.

Of the two, the class-balanced run is the more instructive, precisely because its aggregate score conceals almost everything of interest. On the quadratic score it gains +0.00143 over the baseline, a difference small enough that a reader working from the headline number alone would reasonably conclude that re-weighting had changed very little. The decomposition tells another story entirely: within superclasses the prior-transported channel improves by +0.19713 while instance alignment deteriorates by −0.19569 (95% CI [−0.20728, −0.18411]), so the near-zero total is not a small effect but two large ones that happen to cancel. What re-weighting bought was a much closer fit to the within-superclass frequency distribution, and what it paid for that was per-instance discrimination, a trade the ten-point accuracy drop confirms independently. Even the apparent paradox of a proper score that fails to fall with accuracy has a straightforward explanation. The baseline is badly overconfident, assigning a mean maximum probability of 0.764 while being correct on only 0.3662 of the test set, so the softer predictions of the re-weighted model recover on calibration roughly what they surrender on alignment.

DRW is the more representative case, since it is an improvement of the kind a benchmark submission would actually claim. Its total gain of +0.06606 divides into +0.04206 of prior-transported gain and +0.02400 of instance alignment (95% CI [+0.01340, +0.03460]; randomization $p = .002$). Roughly two thirds of what presents

Table 5: WAGER decomposition on the 10,000-image balanced CIFAR-100-LT test split. Gains use the quadratic score; CI is the Hájek interval on $\widehat{\Delta R}$; p is the 499-shift randomization diagnostic, one-sided for positive alignment, so the value of 1.000 on the CB rows records an alignment gain at the extreme negative end of its randomization distribution.

Comparison	Audit cell ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
CB vs CE	superclass (20)	0.00143	0.19713	-0.19569 [-0.20728, -0.18411]	1.000
CB vs CE	global (1)	0.00143	0.25573	-0.25430 [-0.26801, -0.24058]	1.000
CB vs CE	tier (3)	0.00143	0.25247	-0.25103 [-0.26453, -0.23754]	1.000
DRW vs CE	superclass (20)	0.06606	0.04206	0.02400 [0.01340, 0.03460]	.002
DRW vs CE	global (1)	0.06606	0.03522	0.03085 [0.01860, 0.04309]	.002
DRW vs CE	tier (3)	0.06606	0.03702	0.02904 [0.01702, 0.04107]	.002

Table 6: Alignment gain by training-frequency tier at the primary superclass audit. The deferred schedule concentrates its genuine per-instance improvement on rare classes; both methods lose alignment on the head.

Tier	n	CB vs CE	DRW vs CE	
		$\widehat{\Delta R}$	$\widehat{\Delta T}$	$\widehat{\Delta R}$
Few-shot (< 20)	3000	-0.03337	0.13785	0.09009
Medium-shot (20–100)	3500	-0.11075	0.10799	0.06760
Many-shot (> 100)	3500	-0.41977	-0.03740	-0.07624

itself as a uniform improvement is therefore recoverable from the class-frequency prior alone, while the remaining third is a genuine and statistically significant gain in per-instance discrimination, so deferring the re-weighting evidently does more than re-rank the classes. Neither an accuracy difference nor an aggregate score can make that separation, which is the practical argument for reporting the two channels instead of their sum.

Where the alignment gain sits is visible in Table 6 and Figure 7. For DRW it falls where the long-tail literature intends it to, reaching +0.09009 on few-shot classes and +0.06760 on medium-shot ones, while many-shot classes lose -0.07624 as the schedule shifts capacity away from the head. The class-balanced run reproduces that same ordering with every level displaced downward, its many-shot alignment collapsing to -0.41977. What the decomposition supplies, then, is not only how much of a long-tail gain is real but where in the frequency spectrum it was earned.

These numbers happen to provide an external check on the theory as well. Computing the alignment term directly from the covariance identity of Eq. (7), through a code path that shares nothing with the transport estimator, gives -0.19530; multiplying by the factor $n_c/(n_c - 1)$ that Proposition 2 predicts, at the $n_c = 500$ of a balanced superclass cell, returns -0.195693 against the estimator’s -0.19569. Theorem 1 and Proposition 2 are in that sense confirmed on real trained models, and not only on the synthetic populations used in the unit tests.

Coarsening behaves as Proposition 1 predicts. Moving from the 20 superclass cells to a single global cell changes the credited alignment from -0.19569 to -0.25430; the difference is the between-cell covariance term, which is negative here, so the coarse

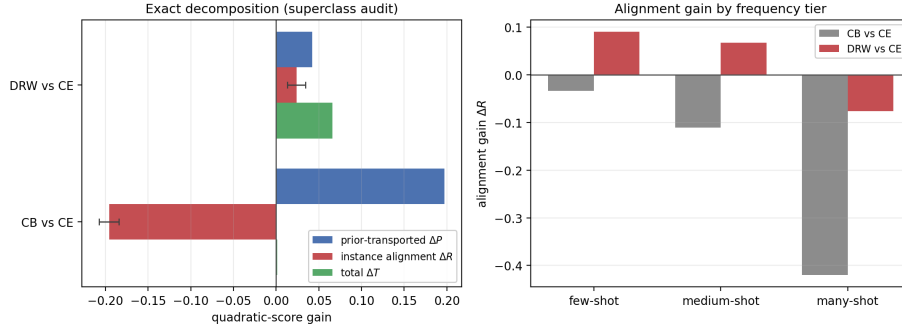


Figure 7: **The same attribution transfers to a second task.** Left: exact decomposition at the superclass audit. The CB total gain (green) is visually almost absent at $+0.00143$ while its two channels reach ± 0.2 – a near-zero aggregate score concealing two large opposing effects. DRW’s smaller total is genuinely split between both channels. Error bars are 95% Hájek intervals on $\widehat{\Delta R}$. Right: the alignment channel by training-frequency tier, showing that DRW’s genuine per-instance gain is concentrated on rare classes and negative on the head.

audit overstates the alignment loss by attributing to it structure that the superclass partition correctly assigns to ΔP . The finer partition is again the conservative reading, for the same reason as in Section 4.4 and now in a second domain.

5. Discussion and Limitations

5.1. What the decomposition does and does not establish

WAGER changes the unit of analysis from a model to a model-pair gain. This matters scientifically. A class-only network may be a better prior estimator than FREQ, but that fact does not become evidence of instance-specific visual use. Conversely, a model can improve visual alignment while losing enough prior score to obscure the improvement in its aggregate metric. The signed identity $\Delta T = \Delta P + \Delta R$ exposes both cases without a separately fitted reference forecaster.

The interpretation remains conditional on the declared audit feature ϕ . A positive ΔR means the new model aligns its probability changes with the correct example better than the old model among examples sharing ϕ . It does not prove causal reasoning, compositional understanding, or robustness. If an unrecorded shortcut varies inside a cell, WAGER will count its predictive use in ΔR . Applications should therefore choose ϕ before looking at results, justify it from the benchmark’s data generation process, and report finer and coarser alternatives. The observed increase under subject-only cells illustrates Proposition 1: coarsening adds a between-cell covariance term of either sign, so it cannot be assumed to leave the estimate unchanged, and only the finest identified partition isolates alignment gain net of every prior regularity expressible in ϕ . This is why the finest defensible prior feature is the conservative primary choice, not merely an empirical convention.

Exact transport also has a positivity requirement: a cell needs at least two examples. WAGER reports the identified fraction and makes no claim for singletons. Very

fine ϕ can therefore improve confound control while reducing coverage. Approximate matching could extend the method to continuous priors, but would replace the exact identity’s clear intervention with modeling assumptions; we leave that extension open.

The image-clustered confidence interval is asymptotic. The cyclic randomization test is finite-group exact only under its sharper within-cell exchangeability null and treats relations as the randomized units, so it is secondary in the SGG analysis. A future image-level randomization design that preserves several relation-cell margins simultaneously would strengthen finite-sample inference for structured benchmarks.

Finally, WAGER evaluates probability vectors, not only top-1 decisions. This is a feature—the method can detect useful redistribution below the argmax—but it requires released logits or probabilities. The quadratic score is bounded and strictly proper; other proper scores answer scale-dependent variants of the same question and should be declared in advance.

5.2. Broader applicability

Four aligned arrays are all the algorithm requires – two models’ probability vectors, the labels, and the prior-cell identifiers – and nothing in Sections 3.1–3.7 is specific to relation prediction. Section 4.6 puts that generality to the test on long-tailed image classification, where taking the benchmark’s coarse superclasses as ϕ lets the decomposition distinguish a re-weighting schedule whose near-zero aggregate score hides two large cancelling channels from one whose genuine improvement is part prior-refitting and part rare-class alignment. The claim of breadth thereby acquires empirical content instead of resting on analogy. That second domain is also where the constraint on ϕ shows its teeth, since the fine class label cannot serve as an audit cell: it would leave every cell label-homogeneous and drive ΔR to zero by construction, producing an artifact that reads exactly like a null result. Elsewhere the natural choices are easy enough to name – a predeclared question template for VQA, an answer format or prompt family for other multimodal benchmarks – though we have not evaluated them, and we flag them as the obvious next domains rather than claim a coverage we have not earned. Whatever the setting, the scientific claim has to stay precise, because what WAGER measures is new prediction–label alignment within the declared cells and nothing more. Its value lies not in any single ϕ defining reasoning once and for all, but in making every attribution an explicit and reproducible counterfactual tied to a declared nuisance structure, one that the same unmodified construction has now answered sensibly on two structurally different tasks.

6. Conclusion

We introduced WAGER, Within-cell Antisymmetric Gain Evaluation of Reasoning, a direct algorithm for attributing the score improvement between two vision models. By crossing labels among examples with the same prior feature, WAGER constructs the prior-only counterfactual from the evaluated data themselves and obtains an exact identity between total, prior-transported, and instance-alignment gains. The residual is an interpretable U-statistic, equals a conditional covariance gain under the quadratic score, and is computable in $O(NK)$ time. Controlled experiments support null calibration and signal recovery, and on Visual Genome the method assigns a class-only

improvement entirely to the prior channel while identifying the additional alignment that spatial geometry supplies. Substituting a frozen CLIP encoder for that geometry produces the sharpest illustration of why the two channels are worth separating: the resulting model scores and ranks worse by every aggregate measure, yet is shown to align with individual instances significantly better than the model it appears to lose to, its whole deficit lying in the annotation prior. Carried over without modification to long-tailed classification, it goes on to distinguish a re-weighting schedule whose near-zero aggregate score conceals two large cancelling channels from one whose genuine improvement proves to be part prior-refitting and part rare-class alignment, a separation that neither an aggregate score nor an accuracy difference can make. It is this gain-specific mechanism, rather than the familiar combination of a frequency baseline with a generic significance test, that constitutes the paper’s technical contribution.

Data availability

All code, cached model outputs and scripts needed to reproduce every number and figure in this paper are publicly available in a repository whose URL is withheld here to preserve anonymity during review; it is given on the title page and will be stated in the published version.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used a generative AI assistant in order to polish the text and correct grammar. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Appendix A. Proofs

Appendix A.1. Proof of Theorem 1

Condition throughout on $C = c$. Expanding over labels gives

$$\Delta T_c = \sum_y \mathbb{E}[H_X(y) \mathbb{1}\{Y = y\}], \quad \Delta P_c = \sum_y \mathbb{E}[H_X(y)] \Pr(Y = y).$$

Subtracting yields the covariance sum in Eq. (6); the first identity is the definition of ΔR_c . Under the quadratic score,

$$H_X(y) = 2\Delta q_y(X) - \{\|q_1(\cdot | X)\|_2^2 - \|q_0(\cdot | X)\|_2^2\}.$$

The bracketed term is independent of the label index. Its contribution summed over y is $\text{Cov}(B(X), \sum_y \mathbb{1}\{Y = y\}) = \text{Cov}(B(X), 1) = 0$. The remaining terms give Eq. (7). If $H_X(y) = H_c(y)$ almost surely, every covariance is zero. $\square$

Appendix A.2. Proof of Theorem 2

Summing the kernel in Eq. (12) over unordered pairs, every observed term $H_i(y_i)$ occurs in $n_c - 1$ pairs with coefficient $1/2$, while the two crossed terms together enumerate every ordered pair $H_i(y_j)$, $i \neq j$, with coefficient $1/2$. Since $\binom{n_c}{2} = n_c(n_c - 1)/2$,

$$\widehat{\Delta R}_c = \frac{1}{n_c} \sum_i H_i(y_i) - \frac{1}{n_c(n_c - 1)} \sum_{i \neq j} H_i(y_j) = \widehat{\Delta T}_c - \widehat{\Delta P}_c.$$

This is deterministic. Under exchangeable sampling, the average of the symmetric order-two kernel is the standard unbiased U-statistic for its population expectation [20]. $\square$

Appendix A.3. Proof of Proposition 1

Fix y and condition throughout on $\bar{\phi} = \bar{c}$. The law of total covariance applied to the pair $(H_X(y), \mathbb{1}\{Y = y\})$ with the finer conditioning variable C nested inside $\bar{c}$ gives

$$\begin{aligned} \text{Cov}(H_X(y), \mathbb{1}\{Y = y\} | \bar{\phi} = \bar{c}) &= \mathbb{E}[\text{Cov}(H_X(y), \mathbb{1}\{Y = y\} | C) | \bar{\phi} = \bar{c}] \\ &\quad + \text{Cov}(\mathbb{E}[H_X(y) | C], \Pr(Y = y | C) | \bar{\phi} = \bar{c}), \end{aligned}$$

using $\mathbb{E}[\mathbb{1}\{Y = y\} | C] = \Pr(Y = y | C)$. Summing over $y = 1, \dots, K$ and applying the population identity of Theorem 1 to the left side (at $\bar{\phi} = \bar{c}$) and to the first term on the right side (at $C = c$, then averaged over c within $\bar{c}$) yields Eq. (8). $\square$

Appendix A.4. Proof of Proposition 2

Fix a cell c with $n_c \geq 2$ and $i \in c$. By definition of the label count, $\sum_{y=1}^K m_{cy} H_i(y) = \sum_{j \in c} H_i(y_j) = H_i(y_i) + \sum_{j \neq i} H_i(y_j)$, and Eq. (14) gives $\sum_{j \neq i} H_i(y_j) = (n_c - 1) \hat{P}_i$. Hence

$$\tilde{P}_i = \frac{1}{n_c} \sum_y m_{cy} H_i(y) = \frac{H_i(y_i) + (n_c - 1) \hat{P}_i}{n_c}.$$

Substituting $H_i(y_i) = \hat{P}_i + \hat{R}_i$ gives $\tilde{P}_i = \hat{P}_i + n_c^{-1} \hat{R}_i$, and $\tilde{R}_i = H_i(y_i) - \tilde{P}_i = \hat{R}_i - n_c^{-1} \hat{R}_i = (n_c - 1) n_c^{-1} \hat{R}_i$. Averaging $\tilde{R}_i$ within cell c and reweighting by n_c across cells, as in the dataset-level definition of $\widehat{\Delta Z}$, gives $\widehat{\Delta R} = N_*^{-1} \sum_c n_c \cdot \frac{n_c - 1}{n_c} \widehat{\Delta R}_c = N_*^{-1} \sum_c (n_c - 1) \widehat{\Delta R}_c$. $\square$

Appendix B. Efficient influence computation

The implementation does not instantiate all pairs. In addition to Eq. (14), let $G_c(y) = \sum_{j \in c} H_j(y)$ and $O_c = \sum_{j \in c} H_j(y_j)$. The pair-kernel mean for observation i is

$$\bar{A}_i = \frac{1}{2} \left[H_i(y_i) + \frac{O_c - H_i(y_i)}{n_c - 1} - P_i - \frac{G_c(y_i) - H_i(y_i)}{n_c - 1} \right].$$

Therefore both the point estimate and Hájek projection require only label counts, column sums, and matrix–vector products inside each cell. For image clusters g , the implemented variance is

$$\widehat{\text{Var}}(\widehat{\Delta R}) = \frac{G}{G - 1} \frac{1}{N_*^2} \sum_{g=1}^G \left(\sum_{i \in g} \widehat{\psi}_i \right)^2,$$

where G is the number of images and $\widehat{\psi}_i$ is Eq. (16). We use a normal critical value; with 27,955 image clusters, small- G corrections are immaterial.

Appendix C. Randomization algorithm

For each cell, sort examples once and index them $0, \dots, n_c - 1$. Randomization b draws $o_c^{(b)}$ uniformly from this index set and maps label position r to $(r + o_c^{(b)}) \bmod n_c$. Independent offsets across cells define an element of the product of cyclic groups. The cell label histogram is unchanged, so $\sum_y m_{cy} H_i(y)$ in Eq. (14) is precomputed; only the assigned label lookup changes. One randomization is $O(N)$ after the $O(NK)$ setup. The plus-one Monte Carlo correction includes the observed assignment and prevents zero p -values.

Appendix D. Implementation and reproducibility

The NumPy implementation is in `wager/antisymmetric.py`. The primary routine `decompose_gain` validates probability matrices, forms score contrasts, excludes singleton cells, computes all three components, verifies the identity numerically, and returns per-instance transported/alignment values plus intervals. The Visual Genome driver is `experiments/run_sgg_wager.py`; controlled validation and sensitivity checks are in `experiments/antisymmetric_simulation.py` and `experiments/antisymmetric_ablations.py`. All random seeds are fixed. Unit tests cover the decomposition identity, quadratic covariance identity, cell-constant null, singleton reporting, cluster inference, randomization power, and log-score path, plus direct numerical checks of Propositions 2 and 1 (the latter against an explicit discrete population with exact expectations, independent of the estimator’s own code path).

Appendix D.1. Compute environment for Sections 4.5–4.6

Model training and feature extraction for both new experiments ran on a single NVIDIA T4 (15 GB) GPU. The WAGER decomposition itself is pure NumPy and requires no accelerator: it consumes the resulting cached probability arrays, as in Section 4.2. All Python code runs under a pinned environment (Python 3.13).

Matched-subsample models (Section 4.5). The three compared predictors share a fixed, seeded ($\text{seed} = 0$) subsample of 100,000 of the 532,987 training relations, drawn without replacement uniformly at the relation level (not the image level); the full 229,605-relation test split is always used for evaluation. All three use the identical head and optimizer – AdamW, learning rate 10^{-3} , weight decay 10^{-5} , 15 epochs, batch size 4096, hidden sizes (256, 128), on standardized features – so they differ only in their input features.

Subject/object crops are taken from the annotated boxes and CLIP-preprocessed (bicubic resize to 224×224 with CLIP’s published normalization constants); a crop with no valid overlap with the image falls back to a blank crop rather than being dropped, so every relation contributes a feature row. CLIP runs in half precision with a batch size of 512 crops and is never fine-tuned. Two implementation details are worth recording. First, the pretrained OpenAI weights were trained with the QuickGELU activation, so the encoder must be instantiated as ViT-B-32-quickgelu; loading the same weights under the plain ViT-B-32 configuration silently substitutes a standard GELU and degrades every embedding. Second, because an annotated object box participates in several relations, crops are deduplicated by (image, box) before encoding and gathered afterwards: the 659,210 crops implied by the audited relations collapse to 422,143 unique ones, removing 36.0% of the encoder work without changing any feature. Only the images referenced by those relations are retrieved, individually rather than through the two bulk archives; of the 71,990 required, three were unavailable upstream and fall back to a blank crop.

ResNet-32 pair (Section 4.6). CIFAR-100-LT uses the standard exponential-profile construction of Cui et al. [10] at imbalance ratio 100: class c (zero-indexed, $0, \dots, 99$) keeps $n_c = \lfloor 500 \cdot \mu^c \rfloor$ training images, where $\mu = 100^{-1/99}$, so class 0 keeps all 500

images and class 99 keeps 5; the balanced 10,000-image test split is untouched. Both models are a from-scratch, randomly initialized CIFAR ResNet-32 (three stages of five basic residual blocks, 16/32/64 channels) trained for 80 epochs with SGD (momentum 0.9, Nesterov, weight decay 2×10^{-4}), batch size 128, standard crop-and-flip augmentation, and a cosine learning-rate schedule (peak 0.1, five-epoch linear warmup). The only difference between the two runs is the per-example loss weight: uniform for the baseline, and the effective-number class-balanced weight $w_c \propto (1 - \beta)/(1 - \beta^{n_c})$ with $\beta = 0.9999$ for the class-balanced model, renormalized to average to one across classes.